



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance



Features span **customer demographics** (Age, Gender, Location, Subscription), **purchase details** (Item, Category, Amount, Season), and **shopping behavior** (Discount, Frequency, Review Rating, Shipping Type).

3,900 Rows

Transactional records

18 Columns

Key features

37 Missing

Review Rating values

Data Preparation in Python

1

Load & Explore

Imported via `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

2

Clean & Standardize

Imputed missing Review Ratings using median per category; renamed columns to `snake_case`.

3

Feature Engineering

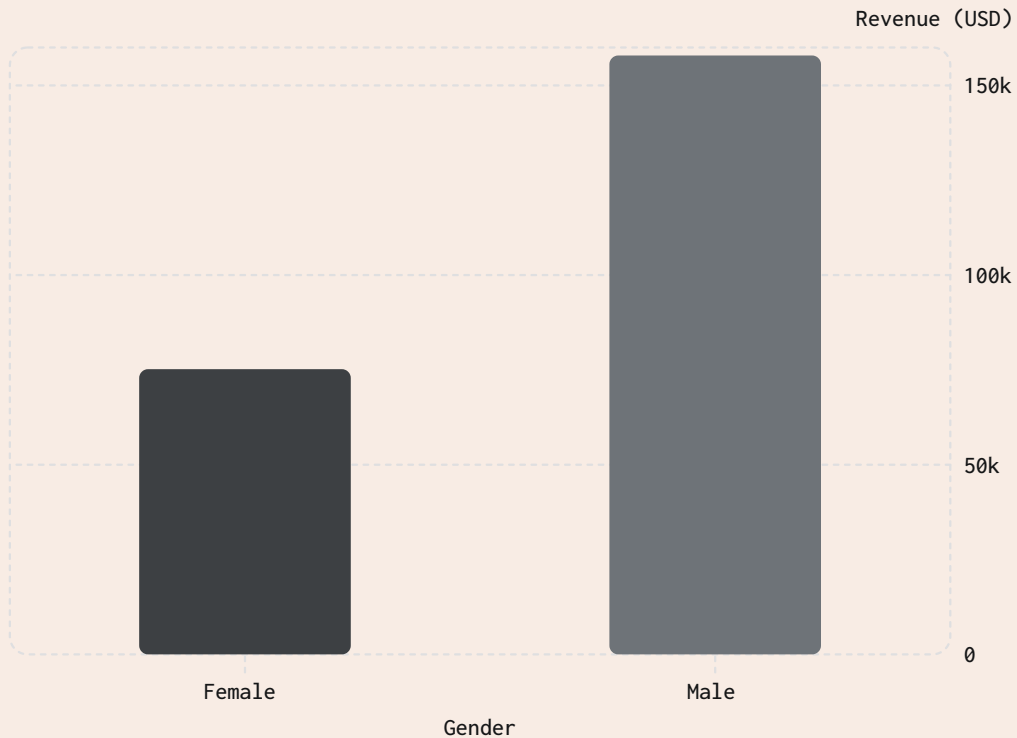
Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

4

Database Integration

Loaded cleaned DataFrame into `PostgreSQL` for SQL analysis.

Revenue & High-Spenders



Revenue by Gender

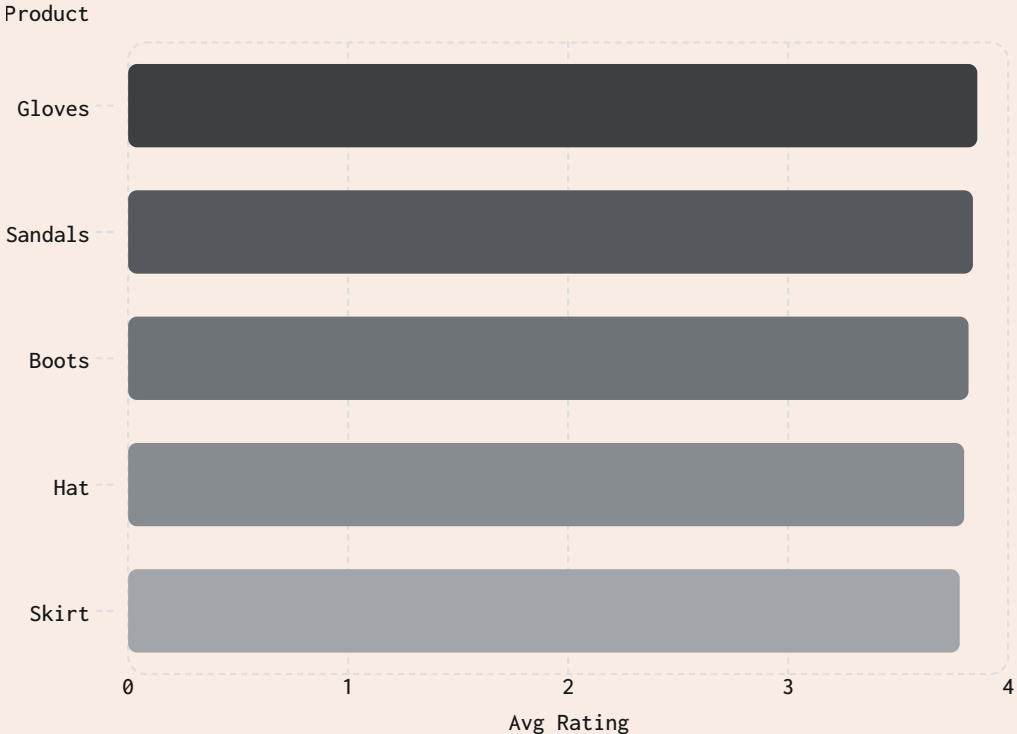
Male customers generated **\$157,890** in revenue vs. **\$75,191** for females – more than double.

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount of **\$59.76**, indicating price-sensitive but high-value shoppers.

Top Products & Shipping Insights

Top 5 Products by Rating



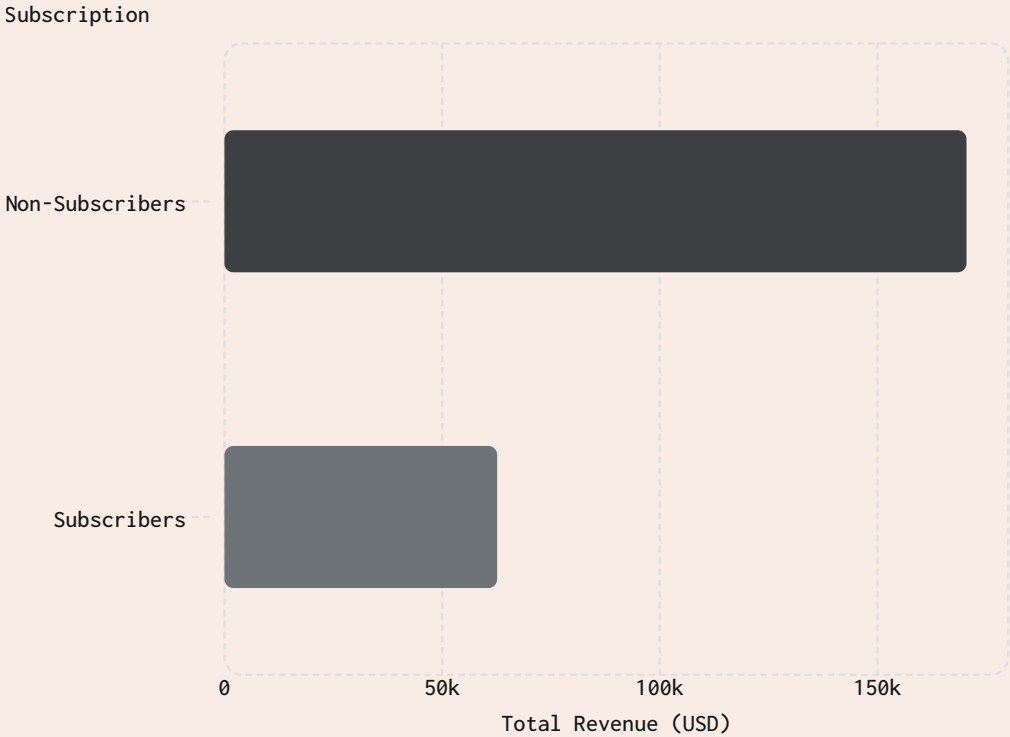
Shipping Type Comparison

Express shipping averages **\$60.48** per order vs. **\$58.46** for Standard – a modest **\$2.02** premium.

Discount-Dependent Products

Hats (50%), Sneakers (49.66%), and Coats (49.07%) have the highest discount rates – nearly half their sales rely on promotions.

Subscribers vs. Non-Subscribers



Despite similar average spend, subscribers represent a smaller, underpenetrated segment – a clear growth opportunity.

Non-Subscribers (73%)

2,847 customers · Avg spend **\$59.87** · Total revenue **\$170,436**

Subscribers (27%)

1,053 customers · Avg spend **\$59.49** · Total revenue **\$62,645**

Customer Segmentation

3,116

Loyal Customers

80% of the base – strong repeat
purchase behavior

701

Returning

18% – potential to convert to
Loyal

83

New Customers

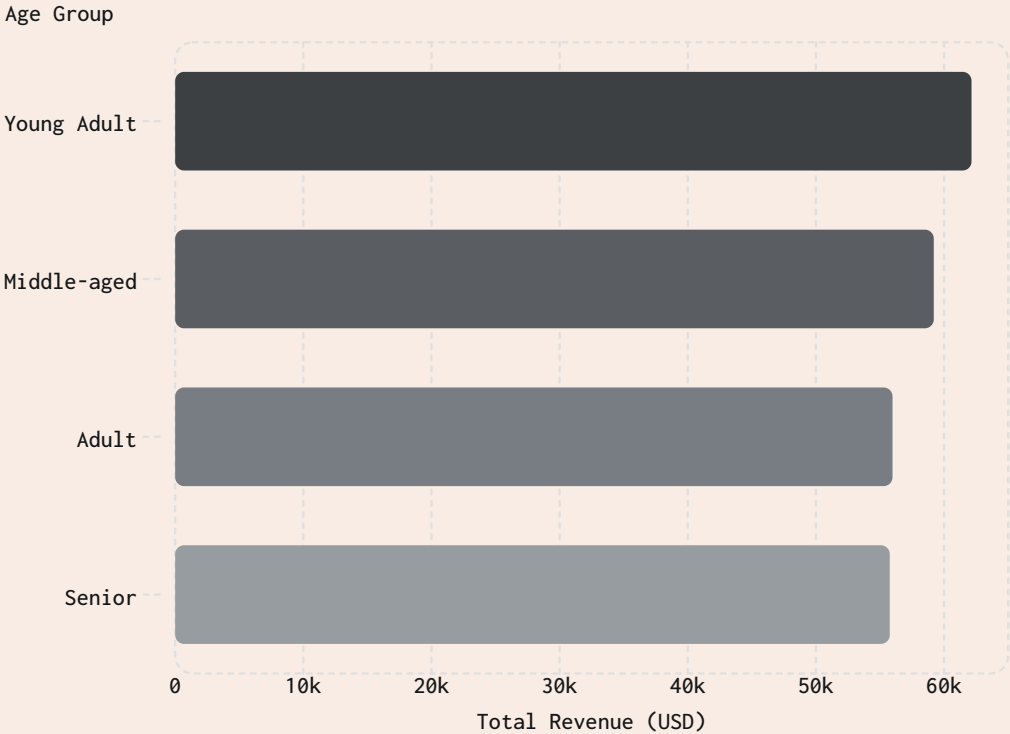
2% – early-stage acquisition
opportunity

Additionally, **958 repeat buyers** (those with >5 purchases) hold
subscriptions vs. 2,518 who don't – suggesting purchase frequency
correlates with subscription likelihood.



Top Products by Category & Age Group Revenue

Revenue by Age Group



Top Products per Category

Category	Top Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

Young Adults lead revenue at **\$62,143**, with all age groups contributing relatively evenly.

Power BI Dashboard

An interactive **Customer Behavior Dashboard** with filters for Subscription Status, Gender, Category, and Shipping Type.

Key Metrics

3.9K customers · Avg purchase
\$59.76 · Avg rating 3.75

Revenue by Category

Clothing \$100K · Accessories
\$75K · Footwear \$30K ·
Outerwear \$15K

Sales by Category

Clothing 1,800 · Accessories 1,200 · Footwear 600 · Outerwear
200



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to grow the 27% base.



Loyalty Programs

Reward repeat buyers to convert them into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control, especially on discount-dependent products.



Targeted Marketing

Focus on high-revenue age groups and express-shipping users.



Product Positioning

Highlight top-rated and best-selling products in campaigns.